

PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Smart Complaint Management System

Campus Facility Complaint Registration, Routing and Resolution Tracking

SOFTWARE TEST PLAN

Item	Details
Group / Course	Team 6 / UE24CS341A — Software Engineering
Document version	2.0 — representative planning set
Product scope	SCMS Release 1.0; six planned Agile sprints
Document status	**Prepared for submission**

Team profile

Member / ownership	SRN / PRN
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1–5. Introduction, Scope and Strategy

1. Purpose / 2. Test items / 3. Features to be tested

Plan functional, security and nonfunctional validation of SCMS against the consolidated SRS and simplified SAD. Test F1 access/admin/reports, F2 intake/evidence/anonymous tracking, F3 routing/allocation/SLA, F4 lifecycle/history/notices/feedback/tracking. All 10 use cases and 57 requirement/rule IDs are in scope via representative cases plus verification methods.

4. Features not to be tested

Physical repairs/emergency response; vendor SMTP internals; AI, native apps, SMS/WhatsApp, ERP/payments, IoT and other excluded scope. Check application mail behaviour with a test backend and later authorized relay test, not third-party infrastructure.

5. Test approach and entry/exit

Level / type	Plan
Unit	pytest checks for forms/validation, routing/SLA/state rules, reports and notice policy.
Integration	Django test client + ORM/storage/mail test backend; role/object/file scope, record consistency and scheduled actions.
System / regression / acceptance	Browser complaint-to-close flows for each role and anonymous reference; rerun affected checks after changes; usability walkthrough.
Security	Authorized isolated-host negative checks: auth/RBAC, CSRF, input/ORM, file access, secrets/debug/TLS/logs and anonymous privacy.
Performance	TC-15 compares controlled measurements to existing SRS targets.

Entry: reviewed baselines, versioned build, synthetic fixtures, isolated environment, storage/default route and test mail/clock available. Exit: agreed execution-ready cases run with actual/evidence records, important requirements verified, critical/high defects resolved, baseline NFR/security criteria met, line/branch coverage report and complete code/test RTM.

5.1 Test volume and stage distinction

This test plan contains 15 representative test cases covering the major functional, security and non-functional requirements. Additional test cases will be expanded during implementation based on the implemented functionality and applicable course requirements.

6–15. Test Management and Traceability

6. Environment / 7. Schedule / 9. Responsibilities

Use Python/Django, pytest/Django test client, coverage, browser, local SQLite and Docker/PostgreSQL test deployment; Django test email backend or controlled SMTP failure fixture. Use synthetic C1/C2, Staff/Heads across D1/D2, A1/A2, active/inactive categories/rules and a controlled clock. Record actual versions/host/commit before execution. Do not test live campus systems without authorization.

Testing follows the six-sprint plan: S1 access/framework; S2 intake/files; S3 anonymous/routing/allocation; S4 lifecycle/history/reassignment; S5 notices/reopen/auto-close; S6 reports/security/performance/regression and evidence. Owners remain F1 Anushka, F2 Atul, F3 Chaitanya, F4 Tanmay; Atul coordinates fixtures/coverage and all cross-review.

8. Deliverables / 10. Risks / 11. Dependencies

Deliver cases/scripts/fixtures, actual execution logs, defects/retests, unit and line/branch coverage output, security/NFR findings, updated RTM and final test summary after execution. Risks are missing build, mail/scheduler availability, unresolved policy/metric definitions and inconsistent environments. Mitigate with an agreed build, controlled fixtures/stubs and explicit decisions.

12. Suspension and resumption

Suspend if environment is unavailable >4 hours, unstable build blocks >30% of cases, or data/security issues make continuation unsafe. Resume after correction/reset, a successful smoke check and recorded team decision. Preserve failure evidence.

13. Case management and traceability

TC-01–15 on pp.4–8 contain preconditions, steps/data and expected outcomes; all Actual Result and Status fields say Not Executed. SRS Appendix C and SCMS_RTM.json map every ID to component/use case/test or verification. TC cases may group related subchecks; they do not prove every clause passed. Expand positive/negative automated checks during implementation.

Method	Planned verification / evidence
VM-01 — Reliability checks	Inject routing/storage failure; repeat/overlap/interruption; restart/container replacement; verify no lost/partial/duplicate records. Not Executed.
VM-02 — Design/code/config review	Review F1–F4 separation, hashing, ORM/escaping/CSRF, object/file scope, anonymous records/logs, secrets/debug, protected history. Pending implementation.
VM-03 — Usability walkthrough	First-time registration/submission within 4 screens; 320 px layout, text statuses, copyable ID, adjacent errors and confirmations. Not Executed.
VM-04 — Build/quality review	Automated positive/negative checks, line/branch reports against baseline targets, SQLite/PostgreSQL runs, Docker build and Jenkins CI/CD/review/static/security evidence. Pending implementation.
VM-05 — Stakeholder review	Confirm category/department values, SLA durations and unresolved policy/metric definitions.

14. Metrics / 15. Approvals

After execution report executed/planned, pass/fail/blocked/not-run, defects/severity, verified requirements, line/branch coverage and measured NFRs. Not-run is not pass. Record tester/date/build/evidence and linked defect per case; reviews/approvals are Pending review.

Representative Tests — TC-01 to TC-03

TC-01 — Registration, login and session controls

Unit / Integration / System

Field	Specification
Preconditions / data	Fresh database; public registration; clock-controlled authentication; test email delivery only. C1 student account; valid password Campus123; short/no-letter/no-digit variants; unknown email.
Steps	1. Register valid C1; retry same email and try invalid passwords/role injection. 2. Log in correctly and with wrong/unknown credentials. 3. Make 5 failed attempts in 15 minutes; retry while locked and after default 15-minute lockout. 4. Log out; repeat after 30 minutes idle.
Expected result	One account with default Complainant role; no self-selected privileges. Duplicate registration gives generic acknowledgement; invalid passwords rejected. Correct login opens role page; failures generic; lockout and logout/idle expiry require fresh authentication.
Actual Result / Status	Not Executed / Not Executed

TC-02 — RBAC and complaint ownership

Integration / Security / System

Field	Specification
Preconditions / data	Two departments D1/D2; C1/C2, Staff S1/S2 and Heads H1/H2; Administrator A1. C1 complaint in D1 assigned S1; other complaint in D2; tampered complaint IDs and role fields.
Steps	1. C2 requests C1 detail/action. 2. S2/H2 request D1 work. 3. Same-department unassigned Staff attempts resolve. 4. C1 calls configuration/report/assignment actions; A1 changes membership then user retries.
Expected result	Cross-user/department requests denied with no state/history change. Unassigned Staff cannot progress/resolve. Only authorized roles act; next request uses current membership. Refusals logged without sensitive data.
Actual Result / Status	Not Executed / Not Executed

TC-03 — Authenticated intake, validation and tracking

Unit / Integration / System

Field	Specification
Preconditions / data	C1 logged in; active Plumbing/Block A/Room 101; default department and storage ready. Title Tap leak; Medium priority; description lengths 19/20/2001; inactive category; unknown location.
Steps	1. Submit missing/invalid fields then valid 20-character description. 2. View acknowledgement, list/filter/detail/history. 3. Retry the same submission after simulated timeout. 4. Check emergency notice and explanatory field errors.
Expected result	Invalid form creates no complaint and preserves safe fields. Valid form yields one SCMS-YYYY-NNNNNN ID, account/time, routing and chronological history. Retry creates no duplicate. Own list filters correctly; emergency notice visible.
Actual Result / Status	Not Executed / Not Executed

Representative Tests — TC-04 to TC-06

TC-04 — Anonymous complaint and reference

Integration / Security / System

Field	Specification
Preconditions / data	No login; active intake settings; anonymous fixture; isolated rate-limit source and controlled clock. Valid complaint, issued reference R1, complaint ID alone, unknown/malformed reference.
Steps	1. Submit anonymously and save one-time R1. 2. Track using R1, complaint ID alone and invalid references. 3. Make 10 attempts in 10 minutes, then an eleventh; retry after window. 4. Inspect record/staff view/export/logs.
Expected result	No registration/login required. Reference is distinct, unguessable, shown once and not recoverable. Valid lookup shows only permitted fields; invalid references have identical no-match. Eleventh lookup throttled. No account/name/email/IP or reference-to-identity linkage; reporter shown anonymous.
Actual Result / Status	Not Executed / Not Executed

TC-05 — Evidence limits and authorized download

Unit / Integration / Security

Field	Specification
Preconditions / data	Valid complaint forms; genuine test JPEG/PNG/PDF; C1 owner and C2 non-owner; controlled storage. 3/4 files; below/above configured 5 MB byte limit; executable renamed image.jpg; genuine JPEG/PNG/PDF.
Steps	1. Submit 3 valid files, then 4, oversized and disguised executable variants. 2. Repeat validation for resolution evidence. 3. Download as owner/authorized staff, then as C2 via direct file reference.
Expected result	Permitted evidence accepted under generated storage names. Invalid count/size/actual type rejects operation; files do not execute. Authorized parent readers may download; direct URL does not bypass scope. Confirm the byte convention before exact-limit execution.
Actual Result / Status	Not Executed / Not Executed

TC-06 — Routing, queues and allocation

Unit / Integration / System

Field	Specification
Preconditions / data	D1/D2/default D0, active/inactive rules, H1 and Staff S1/S2; known complaint fixtures. Category+location, category-only, location-only, no-match rules; equal specificity with different precedence; escalated/priority/age queue records.
Steps	1. Submit each rule combination and unmatched case. 2. Inspect applied rule/default, department/time/ASSIGNED state and queue order. 3. H1 assigns S1 then reassigns with reason; try S2 in another department. 4. Change routing and submit again.
Expected result	Most specific active rule wins, then highest precedence; unmatched complaint reaches D0 with unrouted flag. Queue ordered escalation/priority/age. Same-department allocation records actor/time/reason without state change; cross-department refused. Old complaints retain route; new ones use new rule.
Actual Result / Status	Not Executed / Not Executed

Representative Tests — TC-07 to TC-09

TC-07 — Due dates and scheduled escalation

Unit / Integration / System

Field	Specification
Preconditions / data	Controlled clock; elapsed-hour SLA fixture; max escalation 2 and interval 1 hour are test settings, not campus policy. Open complaint due 1 second ago, future-due complaint, each excluded state; repeated/interrupted run.
Steps	1. Verify category minimum priority and derived due time. 2. Run escalation for past/future-due and excluded states. 3. Repeat within interval, then after interval and at max level. 4. Interrupt/restart run; inspect Head notices/outcomes and prospective configuration change.
Expected result	Only past-due eligible states escalate; one increment per interval up to max, with history/Head notification and no reassignment. Existing due time unchanged after configuration change. Completed work not repeated after interruption; run outcome visible.
Actual Result / Status	Not Executed / Not Executed

TC-08 — Progress, resolution and history

Unit / Integration / System

Field	Specification
Preconditions / data	D1 complaint assigned S1; C1; Head H1; role-authorized fixtures for every state-model row. ASSIGNED/REOPENED/IN_PROGRESS/terminal states; remarks 19/20 characters; invalid ASSIGNED to CLOSED.
Steps	1. Exercise permitted progress/resolve/reject paths from §4.3 and a denied alternative for each rule. 2. Resolve with 19 then 20-character remarks. 3. Attempt concurrent changes and history edit/delete as A1. 4. Try Complainant editing after submission.
Expected result	Only valid role/state actions succeed; short remarks refused, valid resolution recorded. Invalid/competing actions do not corrupt state/history. Actor/time/remarks appended once per accepted action; history uneditable; Complainant post-SUBMITTED edits denied except feedback.
Actual Result / Status	Not Executed / Not Executed

TC-09 — Confirm, rating and reopen

Integration / Security / System

Field	Specification
Preconditions / data	RESOLVED C1-owned and anonymous fixtures; no response-expiry crossing; default max 2 reopens. C1/C2, valid/wrong anonymous reference; ratings 1/5/0/6; rejection reason 19/20 characters.
Steps	1. Confirm as owner with valid/invalid optional rating. 2. Reject with short then valid reason; resolve and repeat through maximum. 3. Attempt third reopen and feedback as C2. 4. Confirm/reject anonymous resolution using valid then wrong reference.
Expected result	Confirmation closes, records actor/time and notifies department; only 1–5 ratings accepted. Valid rejection reopens same department, increments count, keeps history/reasons and notifies Head. Further reopen refused for Head handling. Owner/reference holder only; anonymous feedback grants no extra record/file access.
Actual Result / Status	Not Executed / Not Executed

Representative Tests — TC-10 to TC-12

TC-10 — Auto-close after confirmation period

Unit / Integration / System

Field	Specification
Preconditions / data	RESOLVED, REOPENED and CLOSED fixtures; controlled clock; default 7-day period. Resolution age just below and above 7 days; custom period; repeat scheduler run.
Steps	1. Run before then after confirmation expiry. 2. Repeat with non-RESOLVED state and custom period. 3. Run twice; attempt feedback/closure overlap. Inspect history and closure notification.
Expected result	Only still-RESOLVED expired complaints close as automatic and unconfirmed; earlier/reopened/terminal fixtures unchanged. No duplicate closure or competing invalid outcome. Exact expiry equality is an open boundary decision to confirm before execution.
Actual Result / Status	Not Executed / Not Executed

TC-11 — Notifications and mail failure

Unit / Integration / System

Field	Specification
Preconditions / data	Synthetic role accounts with addresses; anonymous fixture without address; SMTP test backend able to fail. Submission/assignment/start/resolution/escalation/reopen/closure events; unavailable mail; read/unread notices.
Steps	1. Trigger each listed event and inspect recipient/event/channel. 2. View notifications and unread count. 3. Fail SMTP during resolution; exercise up to 3 increasing-delay retries and permanent failure.
Expected result	In-app events match REQ-NOT-01; email uses required subset for addressed parties only. Newest/read/unread behaviour correct. SMTP failure does not undo action or lose in-app notice. Retry/failure recorded; no email to anonymous reporter and no credentials/reference in messages.
Actual Result / Status	Not Executed / Not Executed

TC-12 — Administration safeguards

Unit / Integration / System

Field	Specification
Preconditions / data	A1/A2 Administrators; one referenced category/location/department; D1 Staff/Head; existing complaint. Active/inactive master data; duplicate active rule; durations 0/negative/positive; last Administrator demotion.
Steps	1. Create/edit/deactivate master data; try deleting referenced entry. 2. Change user role/membership and attempt last-Admin removal/self-role change. 3. Create/reorder rules and positive/invalid SLA settings. 4. Compare old/new complaints and audit entries.
Expected result	Administrator-only configuration; no deletion of referenced master data; old references retained. At least one Admin remains; no self-role changes; membership enforced next request. Duplicate active rules and invalid durations refused. Changes audited and routing/SLA changes prospective.
Actual Result / Status	Not Executed / Not Executed

Representative Tests — TC-13 to TC-15

TC-13 — Web security and privacy checks

Integration / Security / System

Field	Specification
Preconditions / data	Authorized isolated test host; C1/C2 role fixtures; protected evidence; deployment configuration available. Missing/invalid CSRF; inert script string; SQL-like text; unknown IDs/reference; forced generic error.
Steps	1. Submit state change without valid CSRF; inspect GET non-mutation. 2. Enter script/SQL-like text in complaint and filters. 3. Attempt unauthorized object/file/report/reference access. 4. Inspect TLS redirects/cookies, SMTP encryption, debug/errors, secrets/logs and anonymous linkage.
Expected result	No unauthorized mutation/data disclosure; text escaped and ORM scope intact. HTTPS and Secure/HttpOnly/SameSite cookies configured; encrypted SMTP. No debug internals, committed secrets, passwords/session/reference logs or anonymous identity links. Record findings.
Actual Result / Status	Not Executed / Not Executed

TC-14 — Report accuracy, department scope and CSV

Integration / System

Field	Specification
Preconditions / data	Frozen synthetic dataset in D1/D2 with known dates, categories/locations, open/overdue/escalated/reopened and resolved cases. A1 global report, H1 D1 report, H2 D2 report; date range; empty range; CSV export.
Steps	1. Compare grouped volumes/state counts/reopen information with fixture records. 2. Calculate mean/median resolution using the agreed formula. 3. Check date range and Head scope. 4. Export CSV and compare displayed figures.
Expected result	Counts match fixtures; Heads cannot see other departments. CSV matches scope/date range/on-screen figures and contains no anonymous identity/reference. Duration formula and date-boundary convention must be agreed before fixed numeric assertions; no result is assumed.
Actual Result / Status	Not Executed / Not Executed

TC-15 — Baseline performance and pagination

Performance / System

Field	Specification
Preconditions / data	Versioned test deployment and synthetic data; record host/database/network and measurement window; authorized load run only. 50 concurrent users; 5,000 open complaints; 12-month report; file up to 5 MB at 2 Mbps; list with >25 records.
Steps	1. Measure each request group server-side under 50-user load. 2. Time 5,000-complaint escalation and 12-month report. 3. Time upload acknowledgement at specified uplink. 4. Inspect list page size.
Expected result	At least 95% within 2s intake/list/detail/tracking and 1.5s queue/actions; escalation <=60s; report <=5s; upload <=30s; <=25 records/request. These are existing SRS targets, not measurements or passed results.
Actual Result / Status	Not Executed / Not Executed